

Deployment Guide: License Plate Recognition API

1. System Requirements

- **OS:** Linux (Recommended) or Windows
- **Runtime:** Python 3.10 (via Conda)
- **System Libraries:** OpenCV requires GL libraries on headless Linux servers.
 - *Ubuntu/Debian command:* `sudo apt-get update && sudo apt-get install ffmpeg libsm6 libxext6 -y`
- **Internet Access:** Required on the **first run** to download the default PaddleOCR models (approx. 20MB).

2. File Structure

Ensure the deployment directory contains the following files:

- `plates_api.py` (The application code)
- `requirements.txt` (Dependencies)
- `plate_model.pt` (**CRITICAL:** This custom YOLO model file must be present in the same directory. It is not included in the standard library.)

3. Installation Steps

A. Set up the Environment

Using Conda with Python 3.10:

```
# 1. Create the environment
conda create -n <name> python=3.10 -y
```

```
# 2. Activate the environment
conda activate <name>
```

B. Install Python Dependencies

Install the required packages using pip (recommended for mixed YOLO/Paddle dependencies):

```
pip install -r requirements.txt
```

4. Running the Service

Use uvicorn to start the server.

```
# Syntax: uvicorn [filename]:[app_instance] --host [ip] --port [port]
uvicorn plates_api:app --host 0.0.0.0 --port 8001
```

- The service will be available at: `http://<SERVER_IP>:<PORT>`

5. API Documentation

Endpoint

POST /api/license-plate

Request Specification

- **Content-Type:** application/json
- **Body:** A JSON object containing the Base64 encoded image string.

Example Payload:

```
{
  "image": "..."
}
```

(Note: The API handles raw Base64 strings or those with the `data: image... prefix`).

Responses

Success (HTTP 200 OK)

The server returns a JSON object. The content depends on detection results:

1. **Plate Detected & Recognized:**

```
JSON
{
  "license_plate": "ABC1234"
}
```

2. **No Plate Detected:**

```
JSON
{
  "message": "No license plate detected"
}
```

3. **Low Confidence (Blurry/Far Image):**

```
JSON
{
  "message": "Model not sure . Provide with a better image (closer more clear)"
}
```

Errors

- **400 Bad Request:** "Invalid base64 image data" or "Decoded data is not a valid image".
- **500 Internal Server Error:** If the server fails to process the temporary file or run the model.

6. Troubleshooting

- **ModuleNotFoundError:** Ensure paddlepaddle is installed (pip install paddlepaddle).
- **FileNotFoundError:** Ensure plate_model.pt is inside the root directory where the command is run.
- **OpenCV Errors:** If you see ImportError: libGL.so.1, install the system libraries mentioned in Section 1.